PROJECT GUIDE

SunDesk Copilot

An agentic RAG support assistant that answers from company documents first, remembers the conversation, and shows its reasoning.

CUSTOMER
Aravalli Solar Works (fictional)

USERS
Service Desk agents and field engineers

USE CASE
Troubleshooting, warranty, installation, billing

BUILT WITH
LangGraph · Groq · Pinecone · Tavily · Streamlit · LangSmith

Aravalli Solar Works is an invented company. Every name, price, error code and policy in the sample knowledge base was written for this project.

1 The Problem

Aravalli Solar Works installs and maintains rooftop solar systems across four service regions. Its support knowledge exists, but it is slow to find.

Service Desk agents and field engineers answer the same questions many times a day:

- What does this error code mean, and what should the customer check first?
- Is this fault covered under warranty?
- How long does installation take, and what must the customer provide?
- Why has the customer's electricity bill not dropped?

The answers are spread across an installation guide, a troubleshooting runbook, a warranty policy, a billing FAQ and a safety handbook. Agents search several documents while the customer waits, and different agents give different answers.

Three approaches that fall short

Keyword search
Returns whole documents instead of one clear answer. The agent still has to read and decide.

A plain chatbot
Answers confidently even when it has no company information, so wrong answers sound right.

Private documents only
Cannot answer questions that depend on current public facts, such as subsidy amounts.

THE CORE BUSINESS CHALLENGE

Staff need fast answers. The company needs those answers to be grounded in its own documents whenever possible, clearly labelled when they are not, and traceable afterwards.

2 Why Simple RAG Is Not Enough

Retrieval-augmented generation (RAG) means finding relevant text first and letting the model answer from it. The simple version has one fixed path, and that is where it breaks.

Simple RAG

- Question, retrieve, generate: always the same path
- Answers even when the retrieved text is unrelated
- Uses the private knowledge base only
- Treats every question as brand new
- No record of why it answered as it did

SunDesk Copilot

- Chooses its next step from what it has found
- Grades the evidence before it writes anything
- Private documents first, web search when needed
- Remembers each chat thread
- Decision trace, audit row and LangSmith run per answer

3 The Solution at a Glance

One workflow, two interfaces, and four outside services, with every answer saved and traced.

6 company documents in the sample knowledge base

35 searchable chunks stored in Pinecone

8 decision steps in the agent workflow

2 interfaces: Streamlit chat and a basic web UI

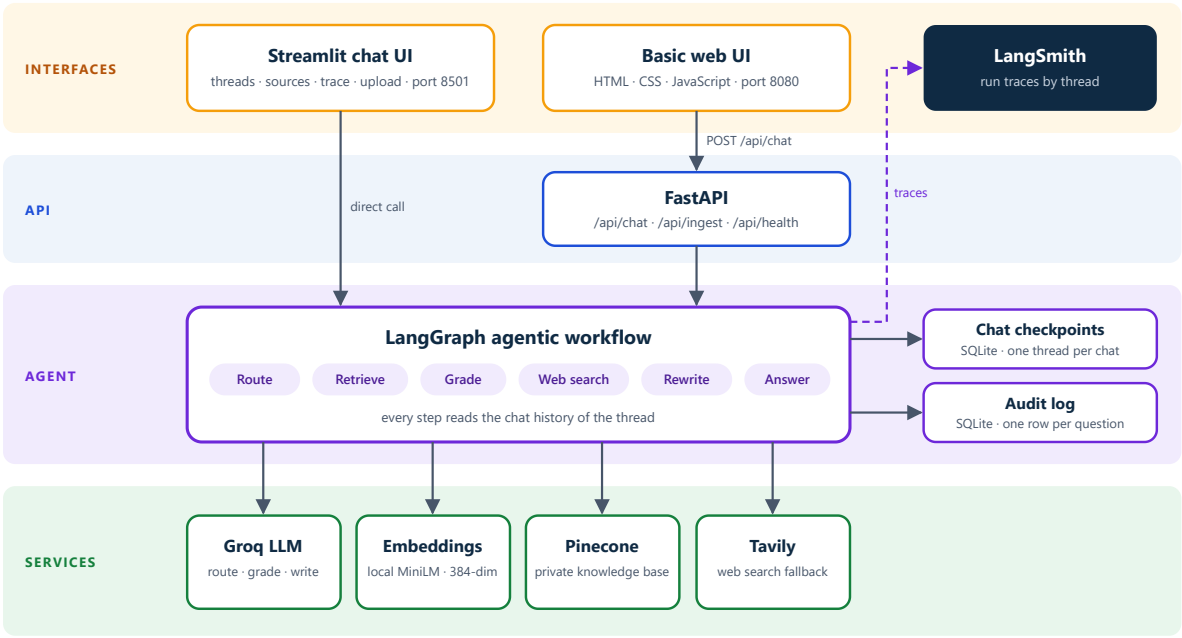


Figure 1. System architecture. Both interfaces drive the same LangGraph workflow. Streamlit calls it directly; the web UI goes through FastAPI.

What each part does

Part	Technology	Job
Agent workflow	LangGraph	Decides the next step from the current state
Language model	Groq, `openai/gpt-oss-20b`	Routes, grades evidence, rewrites queries, writes answers
Embeddings	`all-MiniLM-L6-v2`, runs locally	Turns text into 384 numbers so similar meanings sit close together
Vector database	Pinecone	Stores the private knowledge base and finds the closest chunks
Web search	Tavily	Fallback when the private evidence is weak
Chat memory and audit	SQLite	Saves every thread and logs every question
Observability	LangSmith	Records every step and model call, grouped by thread

4 The Agentic Workflow

"Agentic" means the system decides what to do next from what it has found so far, instead of following one fixed path.

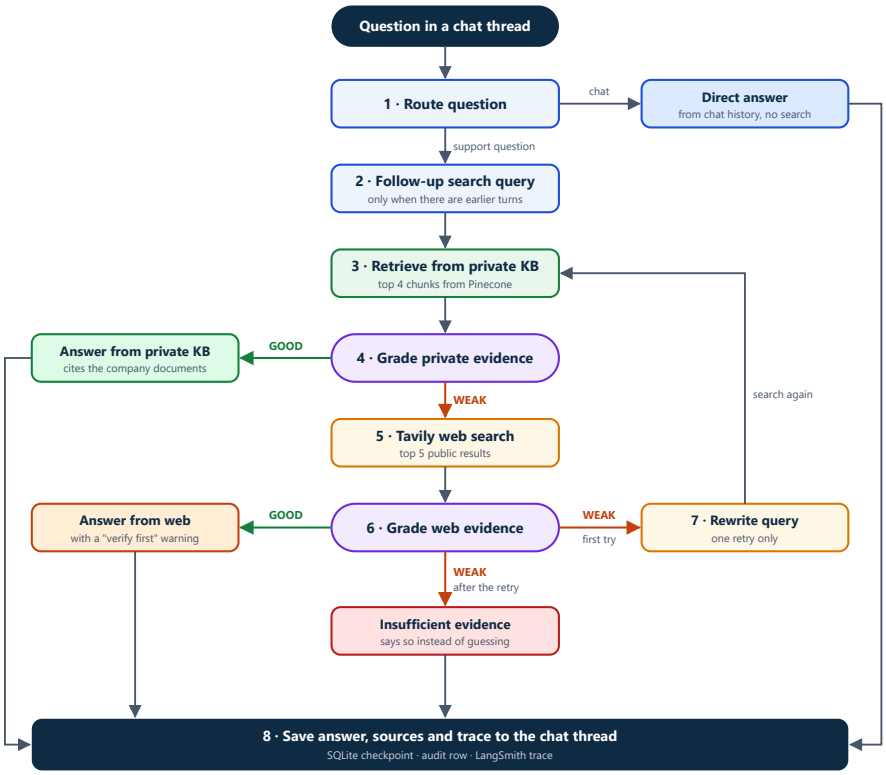


Figure 2. Decision flow for one question. Green arrows follow good evidence, orange arrows follow weak evidence.

Step	What happens	Model call?
1 Route	Decides whether this is a support question or just conversation	Yes
2 Follow-up search query	Rewrites "and for the battery?" into "Battery warranty?" for searching	Only with history
3 Retrieve	Fetches the 4 closest chunks from Pinecone	No
4 Grade private evidence	Judges whether those chunks can answer confidently	Yes
5 Web search	Asks Tavily for public results	No
6 Grade web evidence	Judges whether the web results are relevant	Yes
7 Rewrite query	Adds technical keywords and tries once more	Yes
8 Save	Stores the answer, sources and trace on the thread	No

WHY THE GRADING STEPS MATTER

They are the difference between "I found something" and "I found the answer". Without them, the model would write a fluent reply from whatever was retrieved, relevant or not.

5 The Knowledge Base

The sample knowledge base is six original documents in `docs/knowledge_base/`.

Document	What it covers
Company overview	Service regions, system packages, support channels, ticket priorities P1 to P4, escalation
Installation and commissioning guide	Project timeline, site survey checklist, installation rules, commissioning tests
Inverter troubleshooting runbook	Error codes AS-101 to AS-501, emergency shutdown, no-power and low-generation checks
Warranty and service policy	Warranty periods, exclusions, claims, maintenance plans, refunds
Net metering and billing FAQ	Net-meter process, payment schedule, financing, how subsidies are handled
Safety and maintenance handbook	Panel cleaning, engineer safety rules, maintenance checklist, storm guidance

How a document becomes searchable



Figure 3. Ingestion pipeline. The same pipeline runs for the starter documents and for anything uploaded through the UI.

Why chunks?

A whole document is too long and too mixed to match one question. A 900-character chunk is about one topic, so the right passage can be found and quoted.

Why embeddings?

They let "my gateway is offline" find a passage titled "HelioBox Offline" even though the words differ. The model runs on your own machine, with no API key.

WRITING TIP LEARNED IN THIS PROJECT

Exact codes such as AS-301 are hard for meaning-based search. The runbook states every code in a full sentence ("Error code AS-301 means the inverter is over temperature") so that both formal and casual questions find it.

DELIBERATELY LEFT OUT

Government subsidy amounts and tariff rates change often and differ by state, so the documents do not list them. Those questions are meant to fall through to web search.

6 Conversation Memory

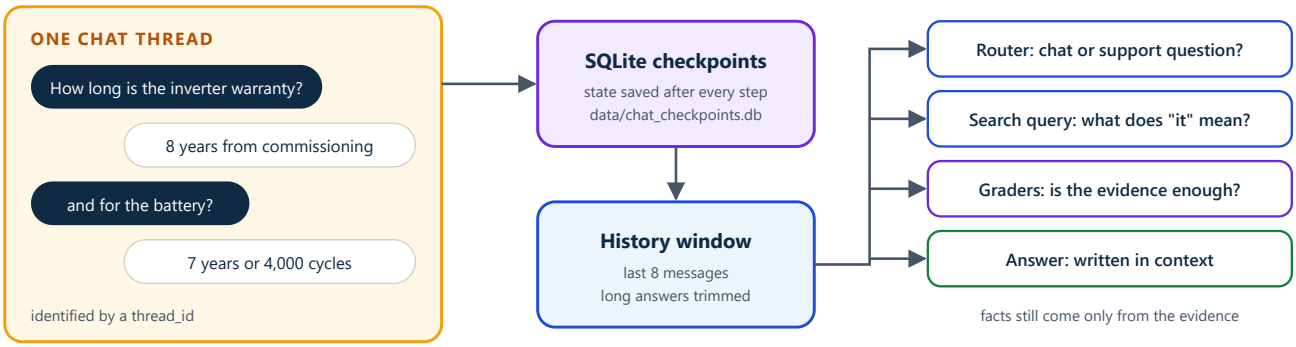


Figure 4. How a chat thread is stored and used.

Each chat has a `thread_id`. After every step LangGraph writes the state of that thread to a SQLite file, so chats survive a restart and can be reopened from the sidebar. The last 8 messages are passed to the router, the search-query step, the graders and the answer step.

> **DESIGN CHOICE**
> Only the search query is rewritten. The user's own words are kept for the answer, and history is used to understand the question, never as a source of facts.

7 Worked Examples

Example 1 · Answer found in the company documents

Private knowledge base

What does error code AS-210 mean?

AS-210 means an earth fault is detected. Carry out the emergency shutdown and book an urgent field visit. Priority P1.

The runbook defines the code, the evidence is graded GOOD, and no web search is made.

Router → KB ▶ Retrieve 4 chunks ▶ KB grade GOOD ▶ Answer from private KB

Example 2 · Private knowledge is not enough

Web search

What is the current rooftop solar subsidy for homes in India?

External web information, verify with the official source before quoting: the central scheme offers a subsidy of up to ...

Subsidy figures are not in the documents. The private evidence is graded WEAK, the web evidence GOOD, and the answer carries a warning.

Router → KB ▶ KB grade WEAK ▶ Tavily search ▶ Web grade GOOD ▶ Answer from web

Example 3 · A follow-up question

Private knowledge base

How long is the inverter warranty?

The inverter is covered for 8 years from the commissioning date.

and for the battery?

The SunVault battery is covered for 7 years or 4,000 cycles, whichever comes first.

The second message makes no sense alone. The thread history turns it into the search query "Battery warranty?".

Router → KB ▶ Follow-up query: Battery warranty? ▶ KB grade GOOD ▶ Answer from private KB

Example 4 · Questions about the conversation

Direct reply

Hi, I'm Shahid. I have a 5 kW SunHome system in Pune.

Hi Shahid! How can I help you with your SunHome system?

Can you summarise what you told me so far in two lines?

A two-line recap written from the chat history.

Neither message needs a document lookup, so no search is made and no search cost is paid.

Router → DIRECT ▶ Direct reply, no retrieval

Example 5 · Nothing reliable found

Insufficient evidence

If the private and the web evidence both stay weak after one rewrite, the system says it could not find reliable evidence and points the user to the Service Desk. It does not guess.

KB grade WEAK ▶ Web grade WEAK ▶ Rewrite query ▶ KB WEAK, Web WEAK ▶ Stopped

8 Seeing What the Agent Did

Three records are kept for every question, each for a different reader.

Agent trace in the UI

For the person asking. A short list of the steps taken, shown under each answer with its sources.

LangSmith run

For the developer. Every step with its prompt, response, timing and token use, grouped by chat thread.

Audit row

For the business. Question, source used and trace, stored in `data/audit.db`.

LangSmith tracing switches on automatically when `LANGSMITH_API_KEY` is present in `.env`. Runs go to the `sundesk-copilot` project.

9 The Interfaces

Streamlit chat UI · port 8501

New chat, saved chat list, delete chat, a coloured source badge, expandable sources and trace per answer, and document upload in the sidebar.

FastAPI and basic web UI · port 8080

A single-page chat with a live trace panel, plus REST endpoints and automatic API documentation at `/docs`.

Endpoint	Purpose
`GET /api/health`	Health check
`POST /api/chat`	Send `question` and an optional `thread_id`; returns answer, source, trace and citations
`POST /api/ingest`	Upload a document; protected by the `X-Admin-Key` header

The colour of the source badge tells the reader how much to trust an answer at a glance:

Private knowledge base · Web search · Direct reply · Insufficient evidence

10 What Was Added Beyond the Starting Template

This project began from a tutorial template for an IT-support copilot.

Area	Change
Domain	New fictional customer, six original documents, new prompts and UI text
Model and embeddings	Groq and local embeddings; no OpenAI key needed
Conversation memory	Chat threads saved as LangGraph checkpoints in SQLite
Use of history	Router, graders and answer steps read earlier turns; follow-ups get a rewritten search query
Streamlit UI	Threaded chat with sources, trace, badges and upload
LangSmith	Automatic tracing with runs tagged by thread
Reliability	Schema-enforced routing and grading, request timeout and retries
API	`/api/chat` accepts an optional `thread_id`

11 Walkthrough

One-time setup

1. **Install.** Create and activate a virtual environment, then run `pip install -r requirements.txt`.
2. **Add keys.** Copy `.env.example` to `.env` and fill in the Groq, Tavily and Pinecone keys. The LangSmith key is optional.
3. **Load the knowledge base.** Run `python ingest_sample_kb.py`. It creates the Pinecone index on first run and uploads the six documents.

> **RUN INGESTION ONCE**
> Running it again adds duplicate chunks. Clear the Pinecone namespace first if you need to reload.

Start the application

Interface	Command	Address
Streamlit chat UI (recommended)	`streamlit run streamlit_app.py`	`http://localhost:8501`
FastAPI and basic web UI	`python run.py`	`http://127.0.0.1:8080`
API documentation	served by FastAPI	`http://127.0.0.1:8080/docs`

Demonstration script

Use the Streamlit UI and open "Agent trace" under each answer.

#	Type this	What to point out
1	What does error code AS-210 mean?	Green badge, runbook cited, no web call
2	What should I check first for it?	The trace shows a follow-up search query naming AS-210
3	How long is the inverter warranty?	Answer from the warranty policy
4	and for the battery?	A short follow-up understood from history
5	What is the current rooftop solar subsidy for homes in India?	KB graded WEAK, orange badge, warning, web links
6	Can you summarise what you told me so far in two lines?	Blue badge, answered from the conversation
7	Click "New chat", then reopen the first chat	The conversation is restored from SQLite
8	Stop and restart Streamlit	Saved chats are still listed
9	Upload a document under "Add company document"	Ask about it straight away; admin key is `change-me` unless set
10	Open the `sundesk-copilot` project in LangSmith	Each question is one run; expand it to see every step

12 Business Value

Shorter handling time
One answer instead of a search through several documents.

Consistent answers
Every agent reads from the same policies and runbooks.

Fewer wasted visits
First checks from the runbook come before a field visit is booked.

Safer answers
Private knowledge preferred, web answers labelled, and honest when it does not know.

Traceability
A decision path, an audit row and a LangSmith run for every answer.

Easy upkeep
New policies are added by uploading a document, not by editing code.

13 Known Limitations and Next Steps

Limitation today	A sensible next step
Groq free-tier rate limits slow answers; one question makes 3 to 9 model calls	Use a paid tier, or merge the routing and query steps into one call
The small model occasionally adds a detail that is not in the documents	Switch to a larger model and add an answer-faithfulness check
Search is by meaning only	Add keyword search alongside it for exact codes and names
Re-running ingestion duplicates chunks	Give each chunk a stable ID so reloads overwrite
One shared admin key, no user login	Add sign-in and per-user chat lists
Web results can be wrong for company-specific terms	Skip web search when the question names an internal code

14 Glossary

Term	Plain meaning
RAG	Retrieval-augmented generation: find relevant text first, then let the model answer from it
Agentic	The system chooses its next step from what it has found, instead of following a fixed path
Chunk	A short passage cut from a document so it can be searched and quoted
Embedding	A list of numbers that represents the meaning of a piece of text
Vector database	A store that finds the passages whose embeddings are closest to the question's
Grounded answer	An answer written only from supplied evidence
Thread	One saved conversation, identified by a `thread_id`
Checkpoint	A saved snapshot of the workflow's state for a thread
Trace	The list of steps the agent took for one question

REBUILDING THIS DOCUMENT

Edit `docs/SunDesk_Copilot_Project_Guide.md`, then run `python docs/build_pdf.py`. The diagrams are SVG files in `docs/assets/` and the styling is in `docs/assets/guide.css`.